

DRAFT: The Scalar-Angular-Torsion Framework: VI. Relativistic Phenomena and the Dark Matter Illusion

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We derive relativistic phenomena and the structural origin of Dark Matter within the Scalar-Angular-Torsion (SAT) framework. By reinterpreting standard Lorentz transformations as emergent consequences of the misalignment angle (θ_4) between 4D worldline filaments and the radial time-flow vector (u^μ), we eliminate the need for axiomatic postulates of relativity. We further define the Obscuration Constant ($\theta_{obs} \approx 0.246$ rad) as the geometric threshold where gauge coupling vanishes, resulting in the “Dark Matter Illusion.” Finally, we predict the Achromatic Phase Snap—a discrete, frequency-independent jump in timing residuals—as the mandatory experimental signature of the discretized 24-cell HSUCV lattice at the critical velocity threshold ($v_{crit} \approx 0.2387c$).

LAY OVERVIEW

In the SAT framework, “moving fast” is reinterpreted as “tilting.” Imagine a 4D filament that usually points straight along the direction of time’s expansion. When an object moves in 3D space, that filament tilts. This tilt, known as the **Misalignment Angle** (θ_4), is the true cause of what we call relativity. **Time Dilation** is not a mysterious stretching of time, but simple mechanical friction. As a filament tilts away from the flow of time, it encounters more drag, causing it to “sink” and trail behind the present moment.

The most startling discovery of this section is the **Dark Matter Illusion**. The universal lattice has a physical “corner” at exactly 14.1 degrees, called the **Obscuration Constant**. When an object’s speed (and thus its tilt) reaches this threshold, the filament hits that corner and precesses out of our observable 3D slice. The matter is still there and still has weight, but it can no longer interact with light. This explains why 80% of the universe seems invisible: it isn’t missing mass, but matter that has tilted past the **Geometric Blackout** threshold. This theory provides a “Smoking Gun” for scientists: at a specific **Critical Velocity** ($v_{crit} \approx 71,500$ km/s), objects will undergo a sudden, discrete “snap” in their timing, proving the universe is built on a rigid, 4D grid.

I. RELATIVISTIC EMERGENCE FROM θ_4

The SAT framework derives standard relativistic effects from the geometric misalignment of 4D filaments relative to the radial time-flow vector u^μ . We reject the axiomatic treatment of Lorentz transformations, reinterpreting them as mechanical consequences of the **Misalignment Angle** (θ_4) and the associated **Deformation Resistance** (κ).

A. Time Dilation as Mechanical Friction

Time Dilation is re-architected as **Projective Resistance** (R). When a filament rotates away from the radial expansion of the S^3 manifold, it ceases to be parallel to the flow of time. This misalignment causes the filament to encounter mechanical friction against the expansion, causing it to “trail” the wavefront of the 3D present. The magnitude of dilation is a direct function of the filament’s orientation:

$$\Delta t' = \Delta t \cdot \sec(\theta_4) \quad (1)$$

where θ_4 represents the angular divergence of the filament tangent from the radial w -axis. This reinterprets the Lorentz factor γ as a purely geometric scaling of filamental drag.

II. THE OBSCURATION CONSTANT AND GEOMETRIC BLACKOUT

A fundamental feature of the 24-cell HSUCV lattice is the **Obscuration Constant** (θ_{obs}). This constant represents the “Geometric Corner” or blackout threshold of the manifold:

$$\theta_{obs} \approx 0.246 \text{ rad } (14.1^\circ) \quad (2)$$

As a worldline’s tangential velocity increases, its internal orientation rotates. Upon reaching the θ_{obs} threshold, the filament encounters a lattice node corner and precesses out of the observable 3D slice.

A. The Dark Matter Illusion

This **Geometric Blackout** provides the structural origin for the “Dark Matter” phenomenon. Matter precessed past θ_{obs} remains physically present in the 4D Zotenwelt and remains gravitationally active through filamental tension. However, because the worldline tangent

has exited the observable gauge sector, it no longer couples to photons. This explains the observed 80% luminosity drop in high-energy celestial bodies without the need for additional non-baryonic particles.

III. CRITICAL VELOCITY AND PHASE SNAPS

The threshold for entering the blackout regime is defined by the **Critical Velocity** (v_{crit}), which is a mandatory product of the Projection Constant B and the speed of light c :

$$v_{crit} = B \cdot c \approx 0.2387c \approx 71,500 \text{ km/s} \quad (3)$$

A. The Achromatic Phase Snap

Because the HSUCV lattice is discretized, worldline rotation is not continuous at high velocities. To main-

tain the Geodesic Requirement and topological stability, the worldline must suddenly “click” or “snap” from one lattice node to the next as it approaches θ_{obs} .

This results in the **Achromatic Phase Snap**: a discrete, frequency-independent jump of exactly 0.246 rad in the timing residuals of relativistic systems (e.g., pulsar glitches). This staccato motion serves as the definitive “Smoking Gun” for the existence of the 4D lattice, as it would be entirely absent in a continuous 3D spacetime.

IV. CONCLUSION OF PART VI

By deriving relativity from filamental misalignment and identifying the structural origin of Dark Matter at the θ_{obs} threshold, the SAT framework provides a unified mechanical explanation for high-energy phenomena. Part VII will focus on the empirical validation of these metrics through the Helium-3 Holotype.